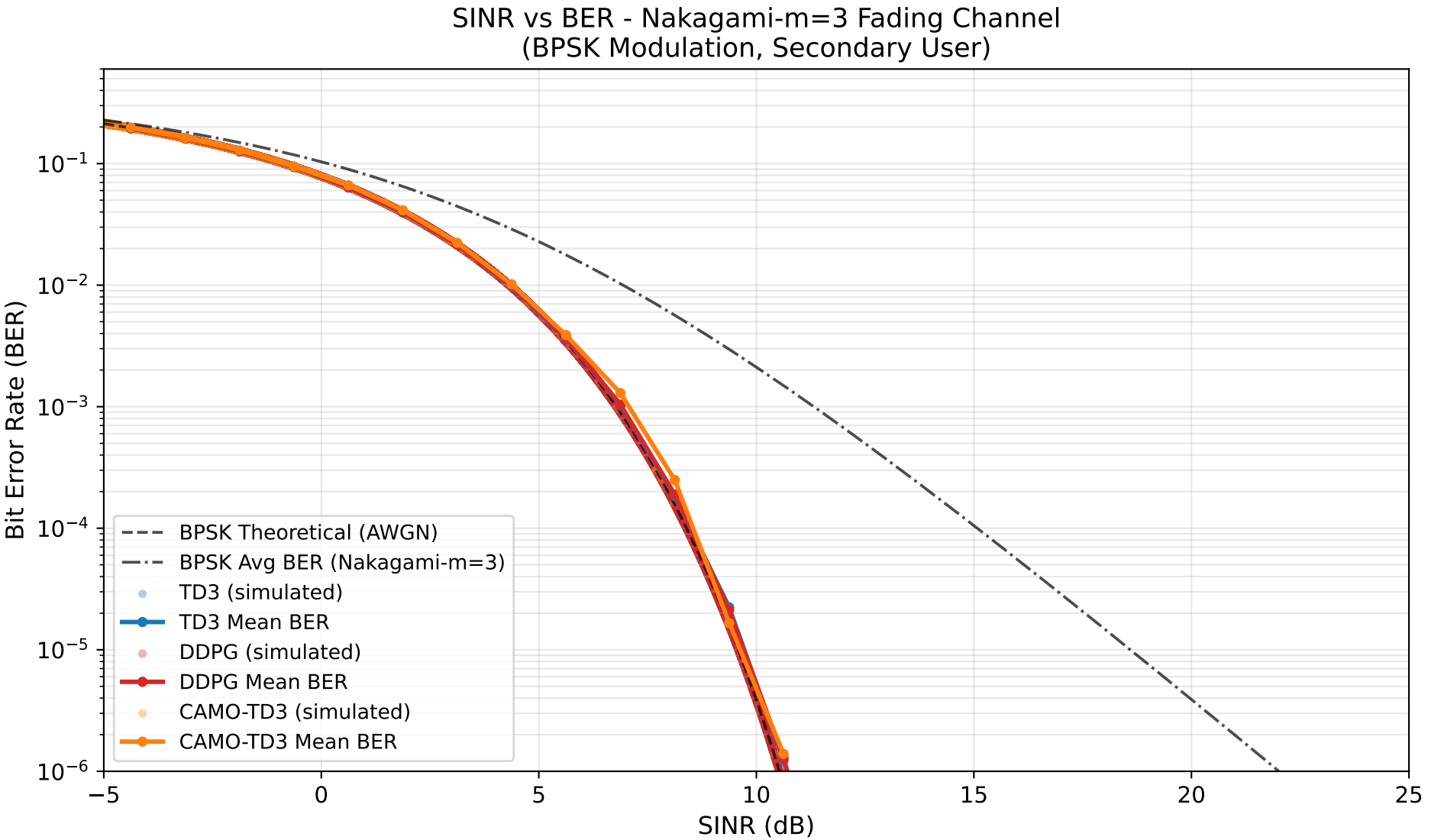
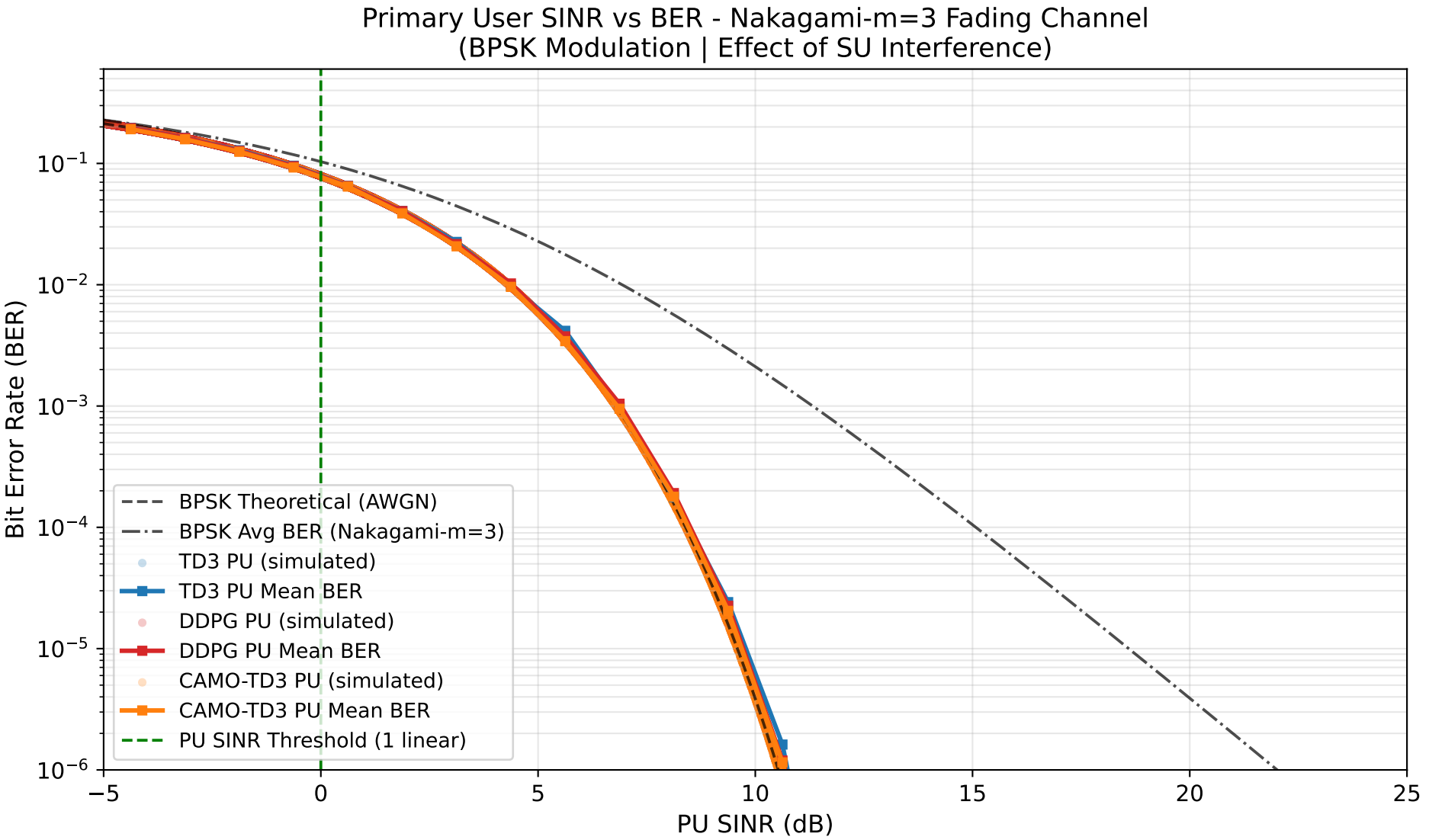
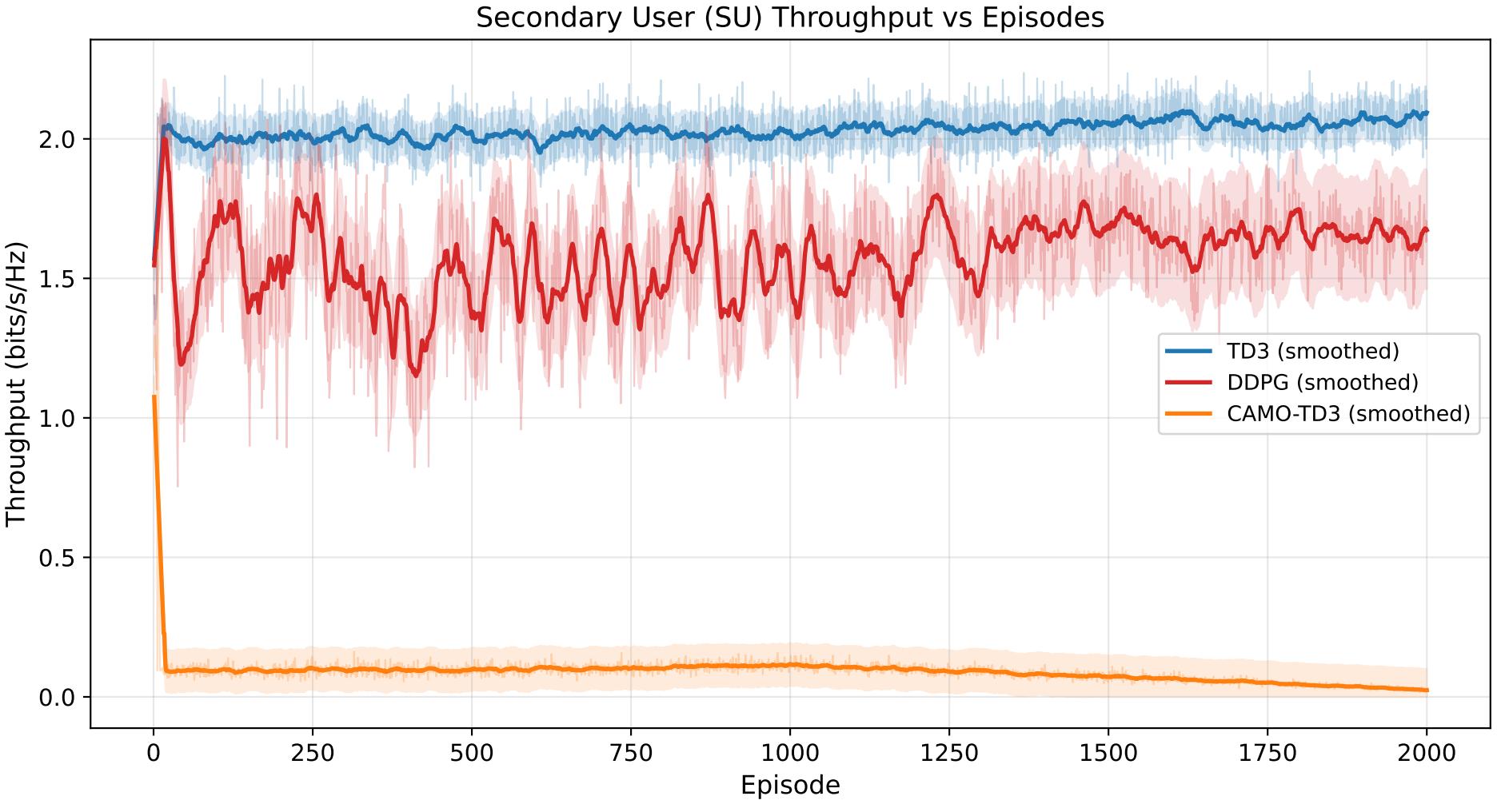
CRN Power Control: TD3 vs DDPG vs CAMO-TD3
Nakagami-m Fading Channel Performance Report

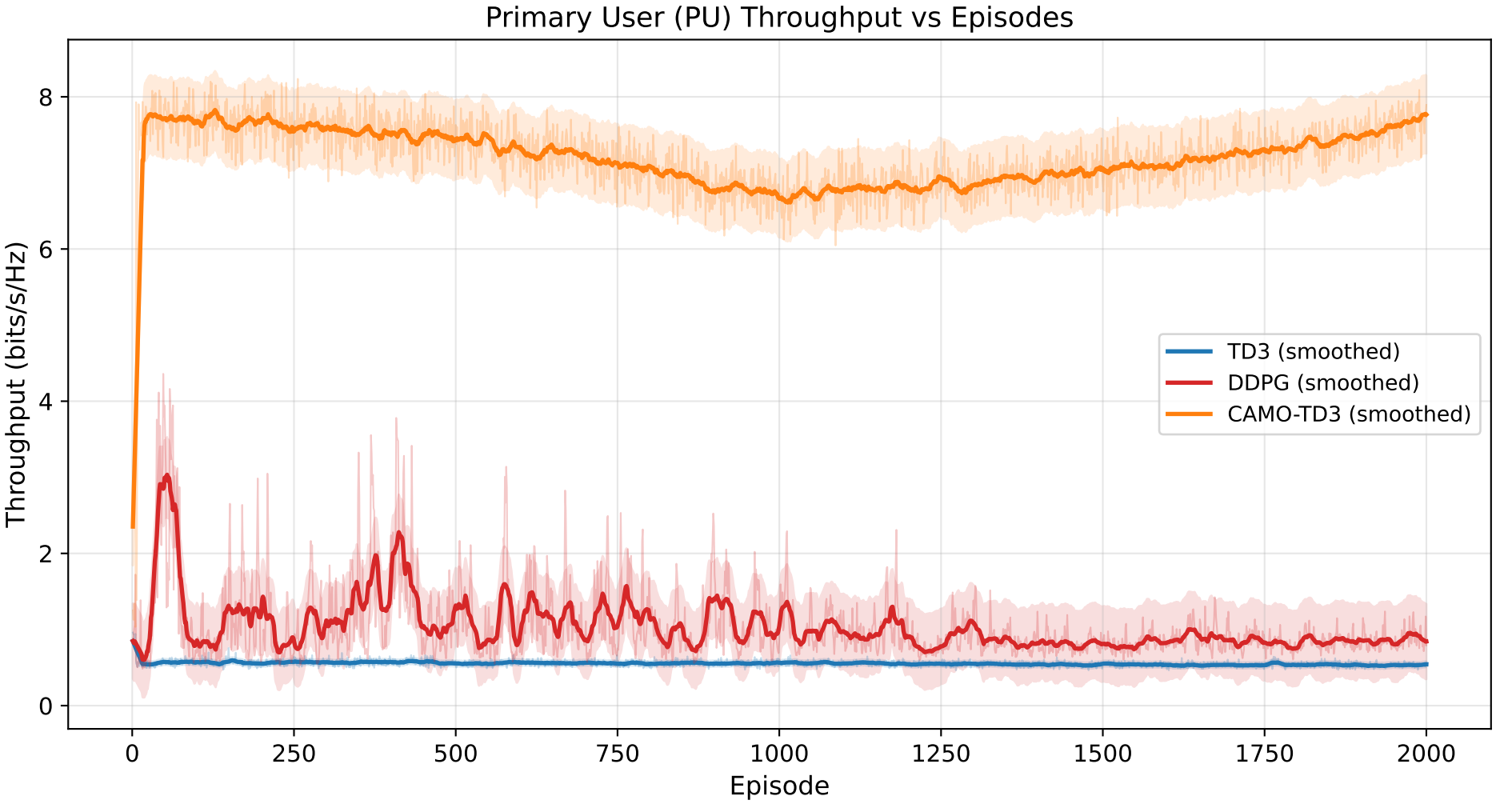
Nakagami-m = 3.0 | Episodes = 2000 | Steps/Episode = 200 | SINR Threshold = 1.0 dB

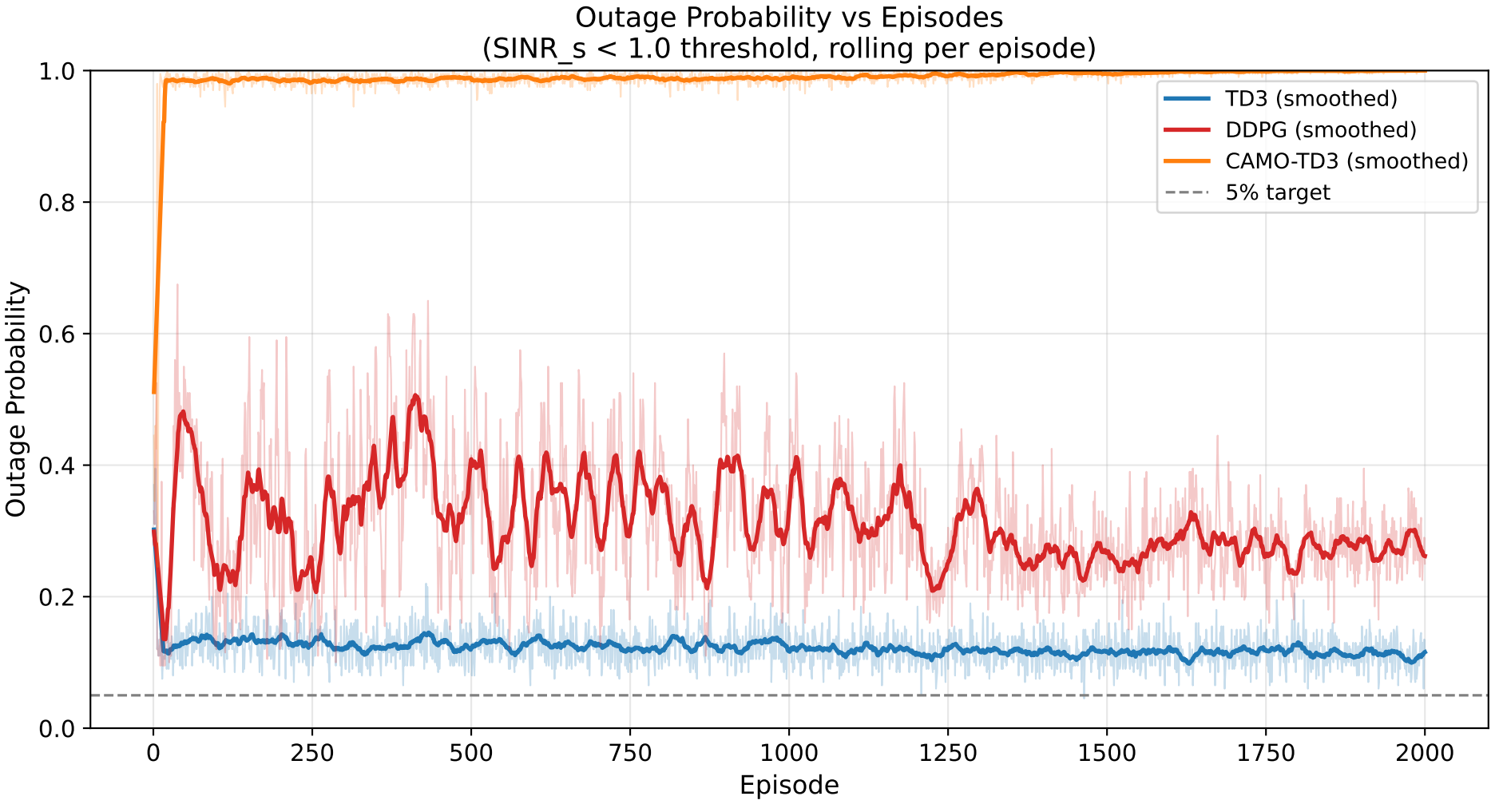
Metric	TD3	DDPG	CAMO-TD3	Winner
Avg Reward (last 100 ep	242.0256	202.8986	5.9055	**TD3**
SU Throughput (bits/s/Hz	2.0719	1.6530	0.0296	**TD3**
PU Throughput (bits/s/Hz	0.5300	0.8724	7.6123	**CAMO-TD3**
Outage Probability	0.1125	0.2783	0.9997	**TD3**
Average BER	0.027771	0.062420	0.452024	**TD3**
Training Time (s)	8913.0	8426.5	20861.8	-

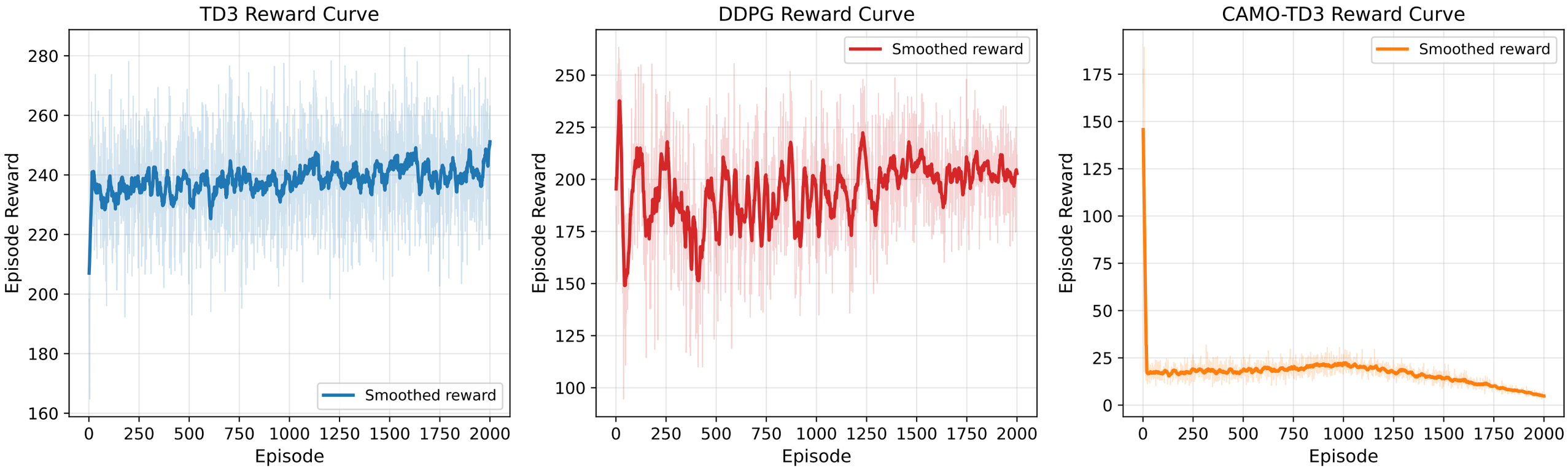

SINR vs BER - Nakagami-m=3 Fading Channel
(BPSK Modulation, Secondary User)
Bit Error Rate (BER)
SINR (dB)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
−5
0
5
10
15
20
25
BPSK Theoretical (AWGN)
BPSK Avg BER (Nakagami-m=3)
TD3 (simulated)
TD3 Mean BER
DDPG (simulated)
DDPG Mean BER
CAMO-TD3 (simulated)
CAMO-TD3 Mean BER

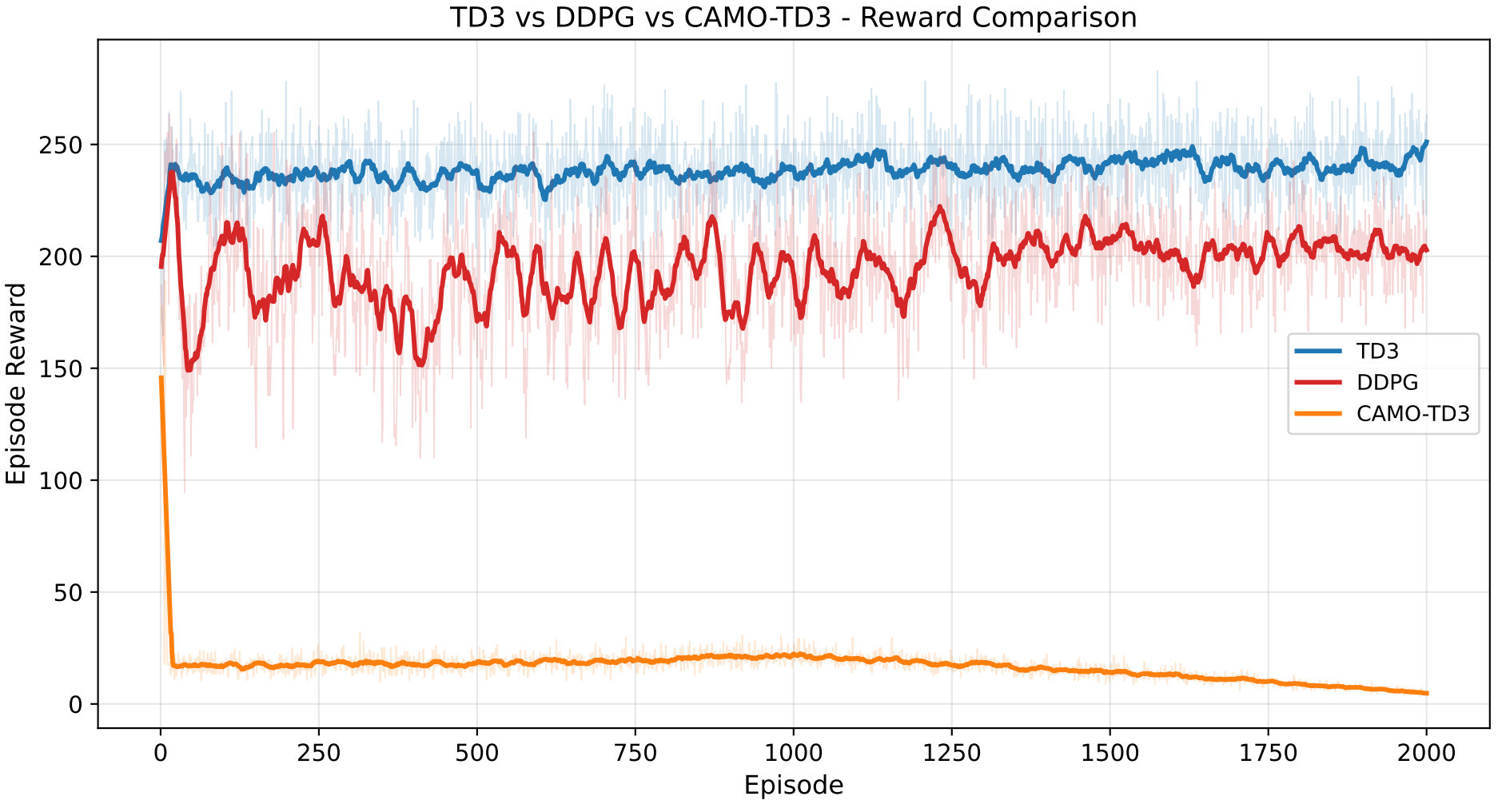

Primary User SINR vs BER - Nakagami-m=3 Fading Channel
(BPSK Modulation | Effect of SU Interference)
Bit Error Rate (BER)
PU SINR (dB)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
−5
0
5
10
15
20
25
BPSK Theoretical (AWGN)
BPSK Avg BER (Nakagami-m=3)
TD3 PU (simulated)
TD3 PU Mean BER
DDPG PU (simulated)
DDPG PU Mean BER
CAMO-TD3 PU (simulated)
CAMO-TD3 PU Mean BER
PU SINR Threshold (1 linear)


Secondary User (SU) Throughput vs Episodes
Throughput (bits/s/Hz)
Episode
0.0
0.5
1.0
1.5
2.0
0
250
500
750
1000
1250
1500
1750
2000
TD3 (smoothed)
DDPG (smoothed)
CAMO-TD3 (smoothed)


Primary User (PU) Throughput vs Episodes
Throughput (bits/s/Hz)
Episode
0
2
4
6
8
0
250
500
750
1000
1250
1500
1750
2000
TD3 (smoothed)
DDPG (smoothed)
CAMO-TD3 (smoothed)


Outage Probability vs Episodes
(SINR_s < 1.0 threshold, rolling per episode)
TD3 (smoothed)
DDPG (smoothed)
CAMO-TD3 (smoothed)
5% target
Outage Probability
Episode
0.0
0.2
0.4
0.6
0.8
1.0
0
250
500
750
1000
1250
1500
1750
2000


Episode Reward Curves - TD3 vs DDPG vs CAMO-TD3
TD3 Reward Curve
Smoothed reward
Episode Reward
Episode
DDPG Reward Curve
Smoothed reward
Episode Reward
Episode
CAMO-TD3 Reward Curve
Smoothed reward
Episode Reward
Episode


TD3 vs DDPG vs CAMO-TD3 - Reward Comparison
Episode Reward
Episode
0
50
100
150
200
250
0
250
500
750
1000
1250
1500
1750
2000
TD3
DDPG
CAMO-TD3